



Backup

Pos1

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Outline

- **What is backup**
- **Backup Level**
- **Backup Plan and Strategy**
- **Backup vs Archive**
- **Backup Media**
- **Backup Retention Policies**
- **Backup Type & Features**
- **Restore Concepts**

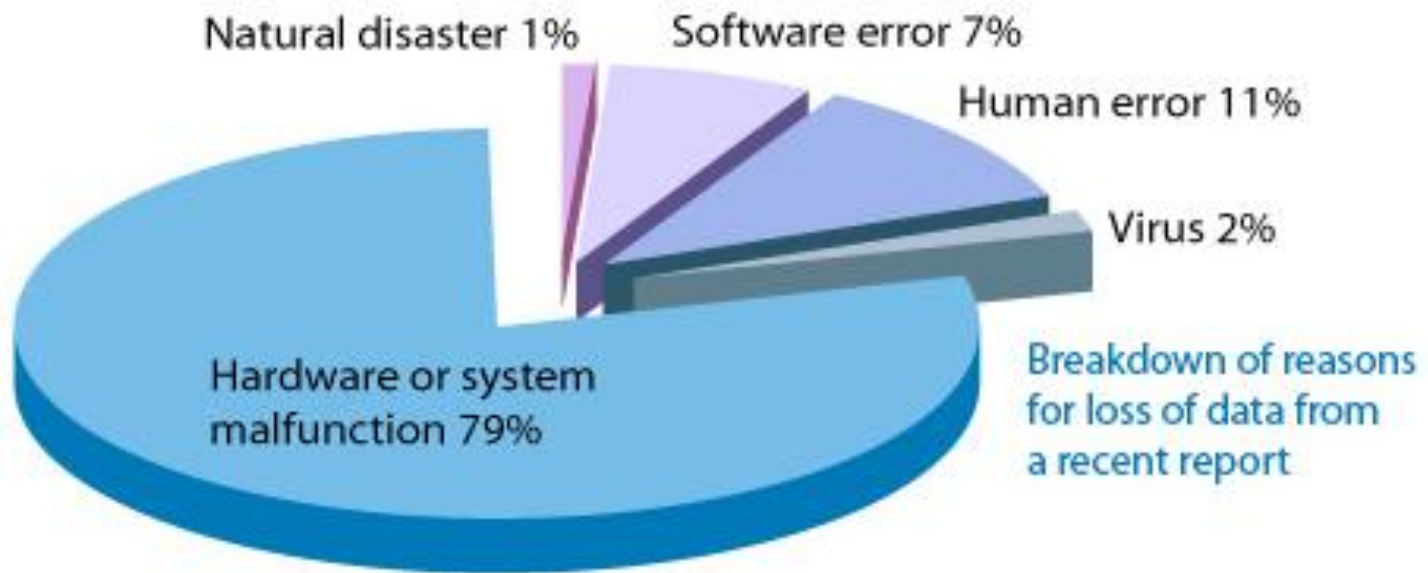
What is backup?

- Backup refers to the process of making copies of data to use in the event the original data are lost or destroyed.
- Backup is a duplicate copy of key information such physical (paper) and computer records.
- A good set of back procedures is necessary to ensure that data is protected.
- Backup may refer to making copies for historical purposes, such as for statistics studies or for historical records or to meet the requirements of a data retention policy.



Why need backup?

- Deletion
- Hardware Failure.
- Corruption
- Lost/Stolen Equipment
- Attach (Ransomware, virus,)
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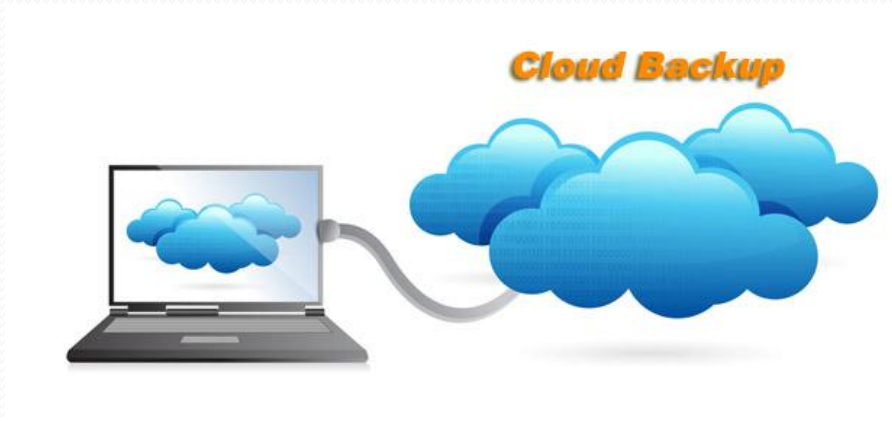
Backup Solution

■ Backup Level:

- ✓ File
- ✓ Application
- ✓ OS
- ✓ Server
- ✓ Storage
- ✓ Cloud
- ✓



- Must take backup from every level because it use in different use case.



Backup Plan and Strategy

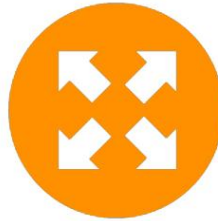
- Backups ensure that your critical data will survive any of the likely hazards. Too many individuals and companies have suffered from having their hardware lost or stolen, destroyed by fires or natural disasters, infected with malware or corrupted by targeted hacker attacks. Common sense dictates that all hard drives will eventually fail; the question is whether or not you are prepared for the worst-case scenario.
- For a one-computer user, the backup strategy can be as simple as copying all important files to another device – or, ideally, several devices – and keeping them in a safe place. However, for multiple computer systems, things can be (and usually are) much more complicated, especially when it comes to virtual environments containing thousands of virtual machines. In these cases, a comprehensive data protection plan should include the **3-2-1 backup rule**.
- This backup strategy entails three components:
 - ✓ 3 copies of your data
 - ✓ 2 local copies on different storage types
 - ✓ 1 backup off-site

Backup Plan and Strategy

3 – 2 – 1 Backup rule



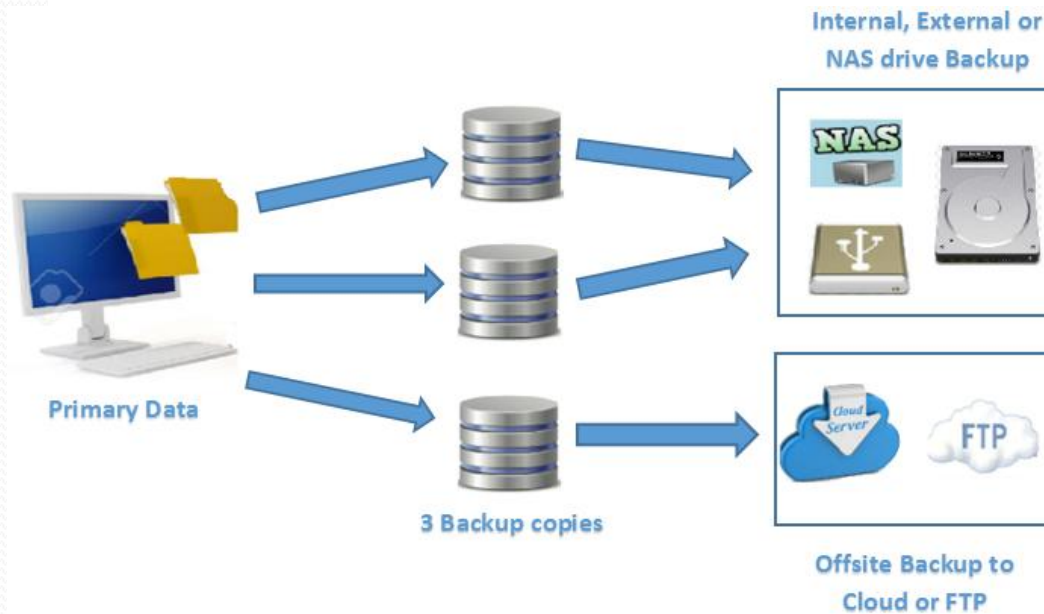
3 copies of
your data



2 different
media



1 copy
off site



- **Note:** 7% of backup restore failed
- What is the different between backup & cluster?

Backup vs Archive

- Organizations need storage products that provide authenticity, long-term retention of data and low total cost of ownership over time, without sacrificing the need for fast access and reliability. However, confusion often arises over the difference between backup and archival storage products and the specific technologies that address each need. With continuing high overall storage growth rates and with approximately 80% of data being static fixed content, companies must understand and distinguish between their backup and archiving needs before choosing the appropriate storage solutions to meet those needs.
- **Backup:** A classic backup application takes regular snapshots of data in order to provide a means of recovering records that have been deleted or destroyed. Most backups are retained only for a few days or weeks as later backup images supersede previous versions. The best way to think of backup is as a short-term insurance policy against an unforeseen disaster; backups help recover information and processes in current use in case they are interrupted, corrupted, or lost.

Backup vs Archive

- **Archive** serve a very different purpose to backups. They preserve inactive information as required by regulations or company policies. An archive is designed to provide fast search and access to years of information and as a result can aid in the discovery of information not currently in use, in case they become useful again to prevent duplicate work or meet an unanticipated regulatory need. In science-driven industries, results and data integrity can be challenged at any time and even inactive documents may need to be retained for many years.

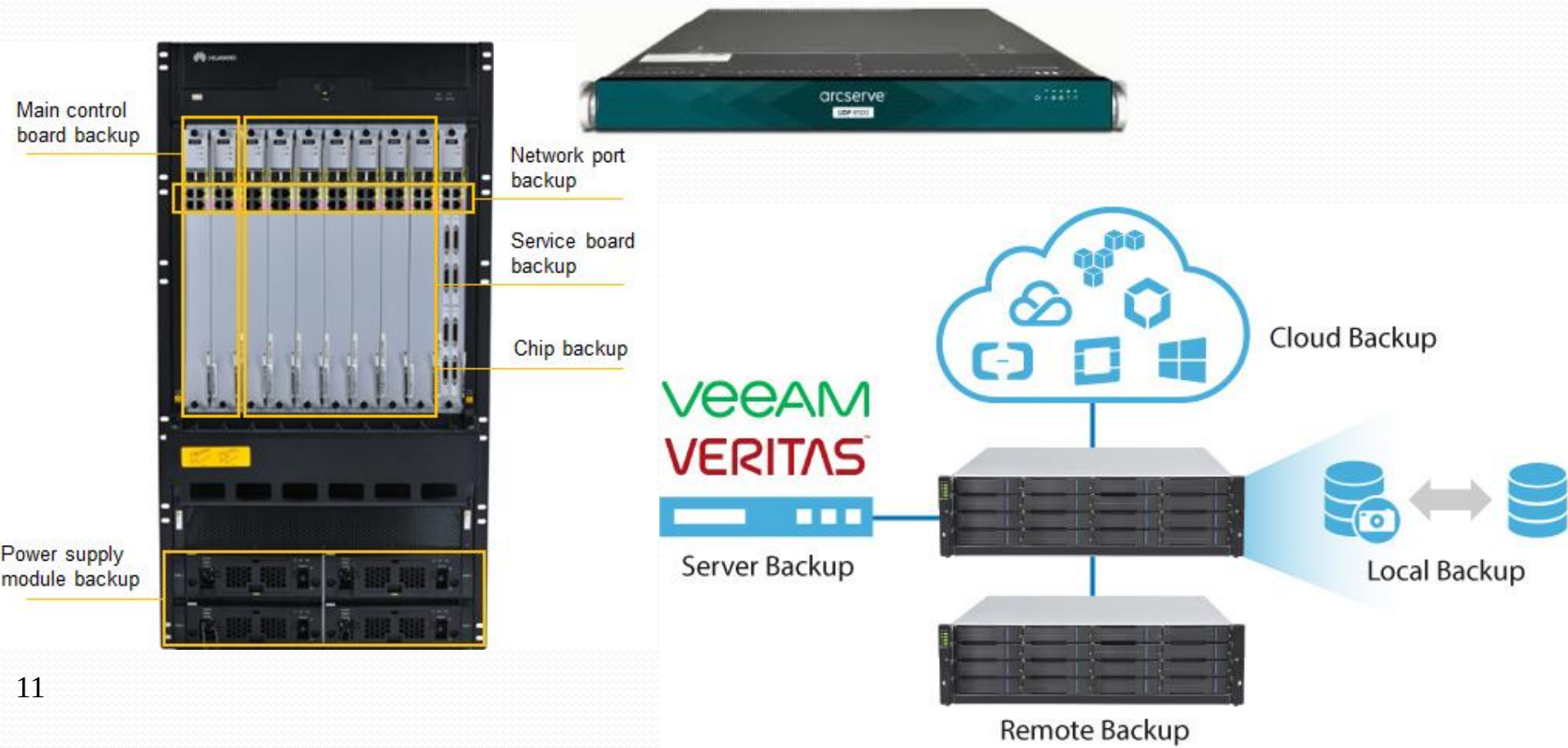
Backup & Archive are Complimentary

- Archiving **Static** data reduces costs
- Backing-up **Active** data is a necessity



Backup Software vs Backup Hardware

- **Backup Software** (VeeAM, VERITAS,)
- **Backup Hardware (Appliance)** (Server with Storages, Processor, RAM, Networks, OS, Applications, RAID configured)



Backup Software vs Backup Hardware

■ Deploy

- يعني اعداد النظام والتطبيقات وتجهيز الـ RAID والشبكة والاعدادات الأخرى وهي أسهل في الـ Appliance عن الـ Software.

■ Performance

- أداء الـ Appliance افضل من الـ Software وهو أهم عامل (النسخة الاحتياطية على نفس جهاز الـ appliance بينما السوفت يكون على جهاز اخر وقد يكون بعيد (سرعة الوصول))

■ Support

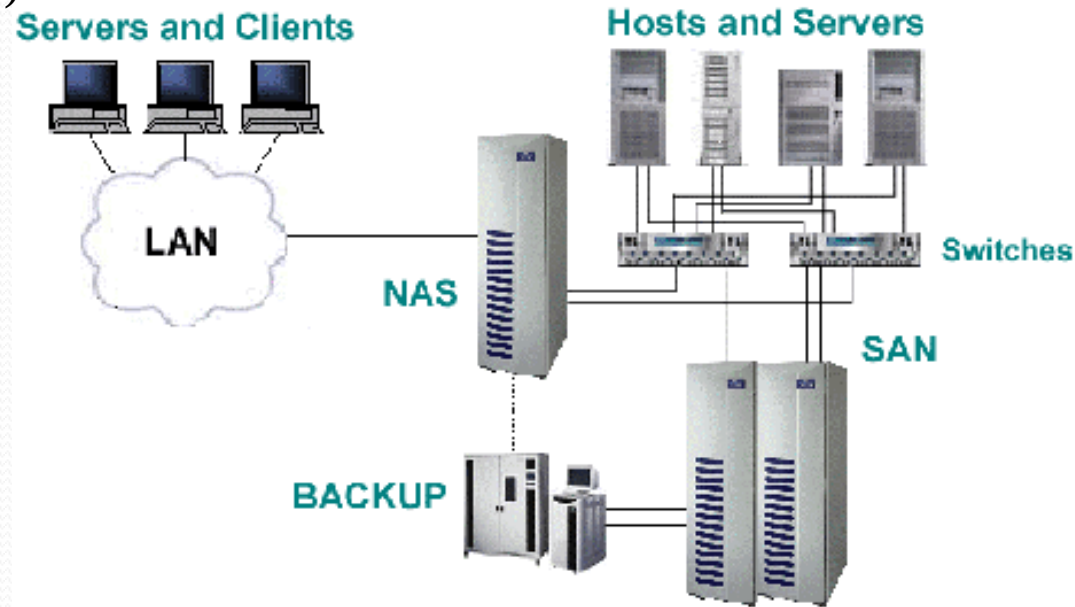
- افضل في الـ Appliance لأنه all-in-one كل المكونات من شركة واحدة أو (one contract support) التوافقية أعلى، والدعم تقدمه شركة واحدة.

■ Cost

- الـ Appliance أغلي ولكن بشكل استراتيجي الـ Software أغلى (بعد تجميع أسعار كل المكونات)
- Software used in small business & Appliance used in middle and large business.
- **Application aware** (Appliance can understand the backup).
- التغييرات التي تحدث اثناء النسخ الاحتياطي يتم اضافتها لاحقا.

Backup Media

- Represent Target or distention.
 - ✓ **Storage** (NAS or SAN use HDD (RAID) or SSD)
 - High Speed & Performance
 - Cannot move away (RAID)
 - Primary
 - Live backup (Daily Backup)



Backup Media

- ✓ **Tape Library** (access Speed & Restore Speed)
 - Can move away
 - Secondary (Long term Retention storage)
 - Archive data or disaster
 - Offline Backup



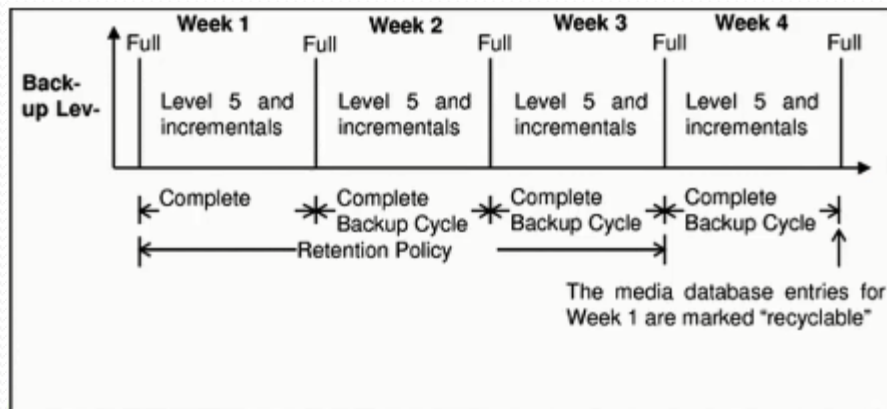
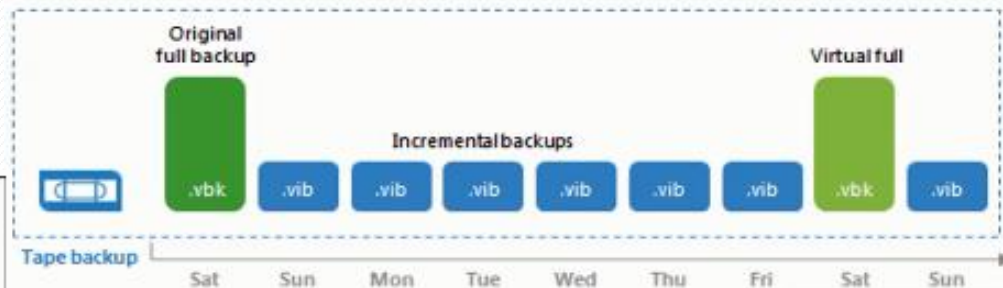
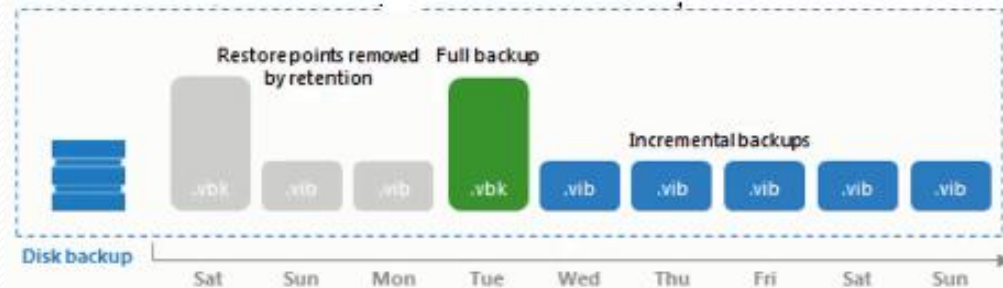
Backup Media

- ✓ **Cloud** (New Technology)
 - Offsite backup
 - Different SaaS
 - Agreement with ISP
 - Cost based size



Retention Policies

- Period of time to save backup in storage capacity (**Short term, Long term**)
 - ✓ Backup must remove after period of time to save storage capacity
 - ✓ Retention Backup **Short term**
 - (Min, Hours, Day, Weeks)
 - Media (Storage)
 - ✓ Retention Backup **Long term**
 - (Months, Years)
 - Tape or Media Backup
 - Based on data important & Agreement with business



Backup Polices and Sheet

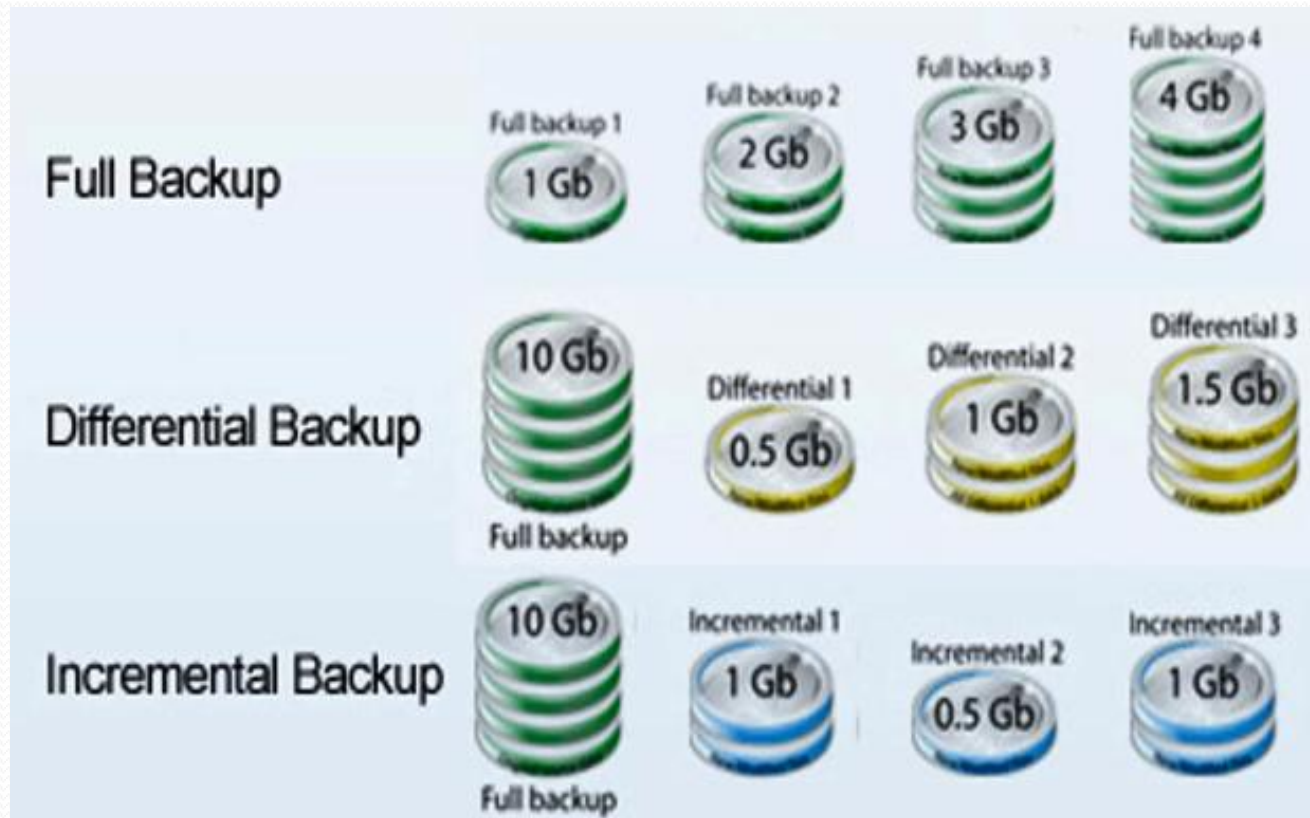
- **Backup Polices and Sheet** (Documented in Excel Sheet)

- ✓ Backup Job name
- ✓ Servers or Application Type
- ✓ Owner
- ✓ Backup Level (App, DB, OS, File)
- ✓ Scheduler (Time)
- ✓ Backup type (Full, Incr, Diff)
- ✓ Backup Media (Storage, Tape, Cloud)
- ✓ Retention Policies
- ✓ Replication if need

[illegible]

Backup Type (Methodologies)

- A type of backup actually defines how the data is copied from the source to the destination.
- Backup granularity: depend on business needs and required RTO/RPO.



Backup Type

- **Backup type:**

- ✓ **Full Backup:** A complete File level zipped copy of source to the destination (Backup of the complete data at a certain point in time)

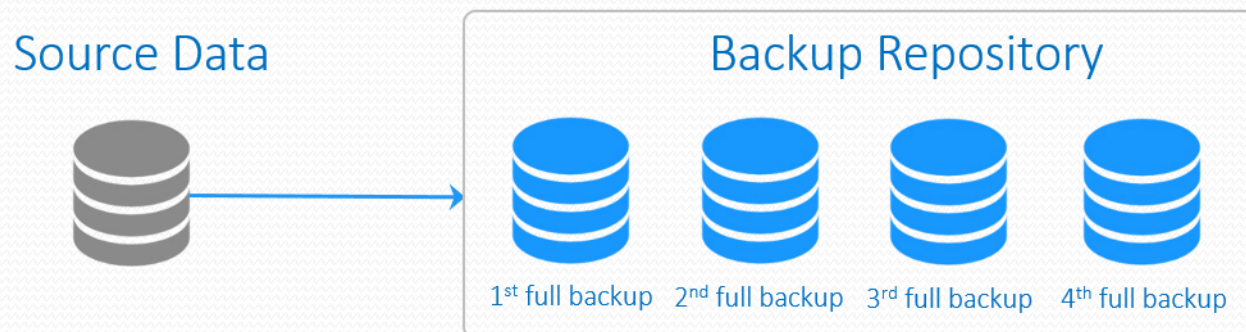
- **Advantages**

- ☐ Fast data recovery in case of disaster
- ☐ Better storage management, since the entire data set is stored in a single backup file

- **Disadvantages**

- ☐ More time consuming
- ☐ Require more space

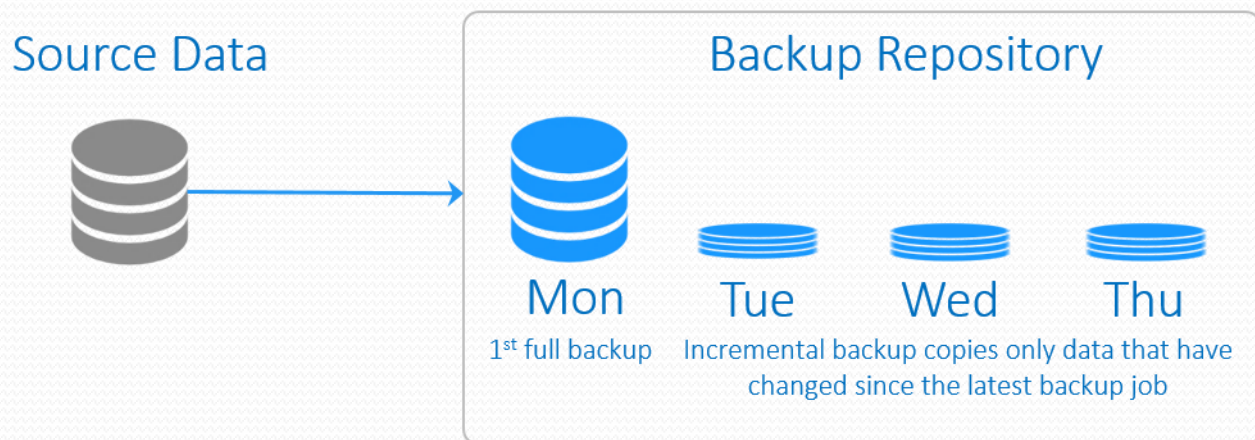
Full Backup



Backup Type

- ✓ **Incremental Backup:** Copies data that has changed since the last full or incremental backup which ever has occurred more recently.
 - **Advantages**
 - ☐ Backup times is faster than full backup
 - ☐ Requires less disk/space
 - **Disadvantages**
 - ☐ In order to restore all the files, you must have all the incremental backups available
 - ☐ Take longer time to restore a specific file.

Incremental Backup

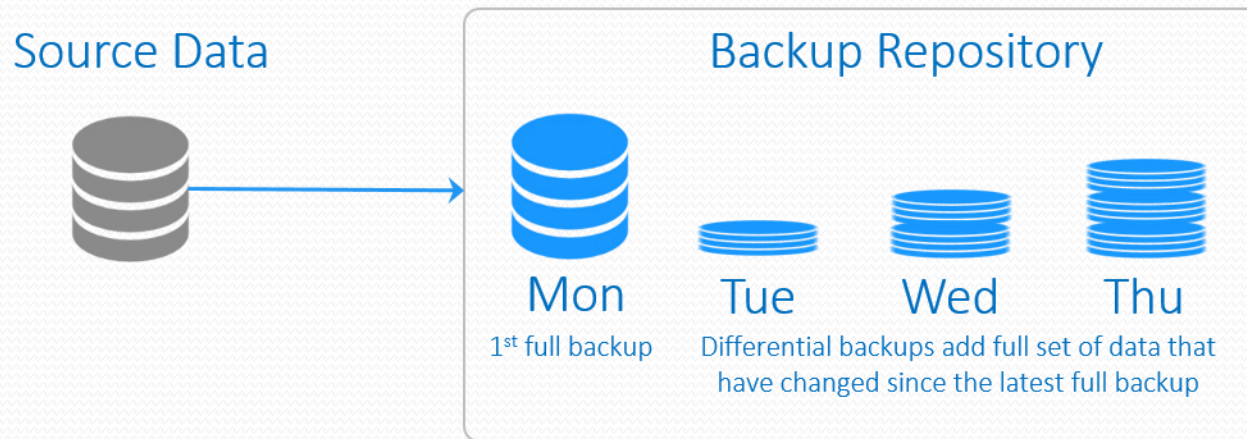


Backup Type

✓ Differential Backup:

- Incremental until new full backup
- Benefits in Restore

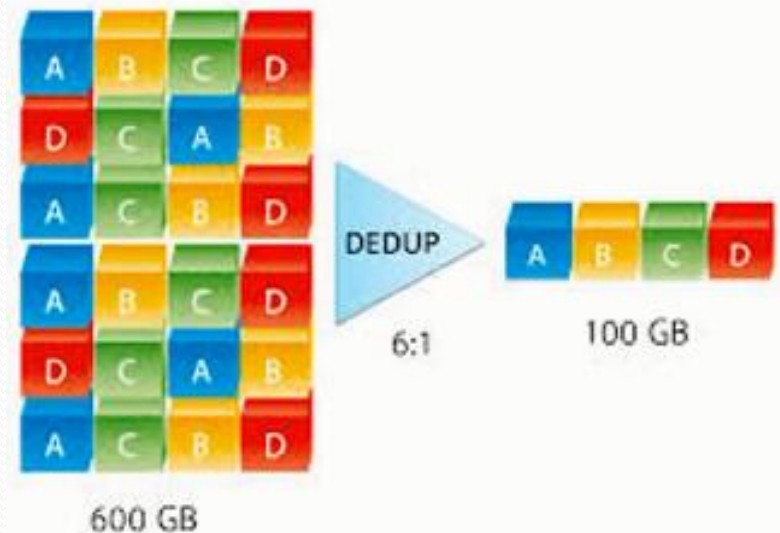
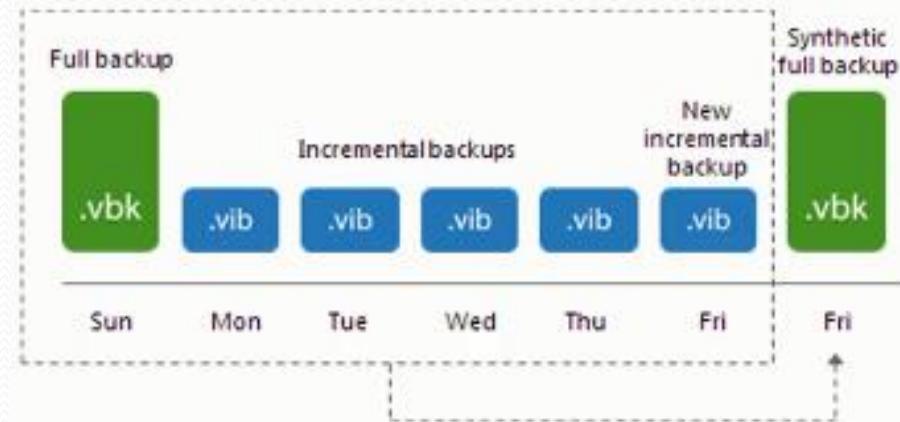
Differential Backup



Backup Features

■ Backup Features

- ✓ Synthetic Backup (Revers incremental, Backup Accelerations)
 - Merge incr. with full without take any new backup copies
 - Best in time, size & performance
- ✓ Compression (saving size, Tape)
- ✓ Deduplication (saving size, Media Storage)
- ✓ Indexing (Speed up the restore process per item)
- ✓ Encryptions
- ✓ Restore by users (based in permission)
- ✓ Reporting (To reach the best performance)



Restore Concepts

■ Restore Concepts

- ✓ Why need Restore? (used in disasters)
 - 7% from Restore not work (must Test Restore)
- ✓ Restore in Different Environment
 - May be there is problem in original
- ✓ Restore Time.
 - Take double time take in backup
 - Important in RTO & RPO
- ✓ Restore from Media
 - Restore from tape take more time than other media (Calculate worst case)
 - Restore from cloud based in connections
- ✓ The way to Restore
 - Based on backup type (full, incr, diff) (most solution now take incr with Synthetic Backup)



Restore Level

- **Restore Level** (understand applications)

- ✓ Item
- ✓ File
- ✓ Folder
- ✓ OS
- ✓ Full
- ✓ VM
- ✓ Application
- ✓ Configuration
- ✓ Email
- ✓ Database (SQL Quarry)
- ✓ Server